

Smart Wearables App: User Guide

This document provides a guide for the *Smart Wearables App*, a Flutter application designed to interface with the Course Development Board running the *"MainBoard_IMU_Logger"* firmware.

The app's primary purpose is to provide a real-time data visualization tool for the IMU sensor data streamed from the board.

In particular, the "Smart Wearables App":

- Scans for BLE devices and connects to the BLE device on the course development board.
- Receives real-time IMU data stream from the onboard IMU sensor.
- Parses raw 20-byte data packets to extract accelerometer X, Y, and Z-axis values.
- Converts raw 16-bit integer data into floating-point 'g' units.
- Plots all three axes on real-time, auto-scrolling line charts.

Section 0: Setup

Software Requirements

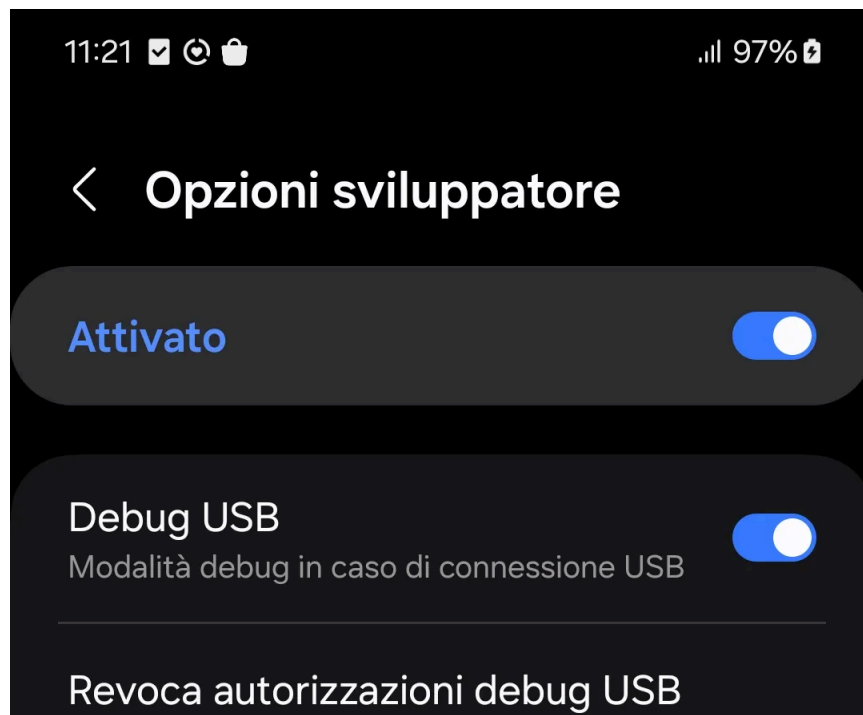
- **Flutter SDK:** <https://docs.flutter.dev/get-started/quick>
- **Visual Studio Code (VS Code):** The recommended code editor. Please refer to the *"How to install VS Code and Python"* guide.
- **Flutter & Dart Plugins:** Must be installed in VS Code from the Extensions Marketplace.
- **Android Studio:** Required to install the Android SDK, command-line tools, and device drivers. <https://developer.android.com/studio?hl=it>

Hardware Requirements

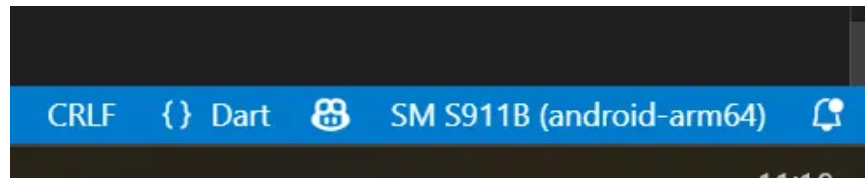
- An Android smartphone for testing.
- The Course Development Board running the "*MainBoard_IMU_Logger* " firmware.

Section 1: How to Run the Project

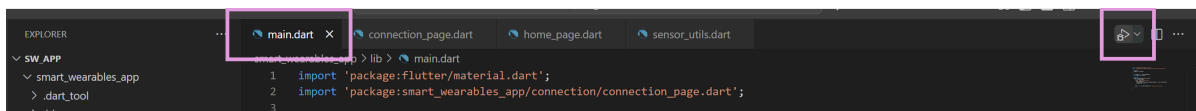
1. Download the project source code "smart_wearables_app".
2. Open the project folder in VS Code.
3. Open the integrated terminal (Terminal > New Terminal).
4. Run `flutter clean` to clean the Flutter project.
5. Run `flutter pub get` to install all required dependencies.
6. Connect your smartphone to your computer via USB (ensure **Developer Mode** and **USB Debugging** are enabled on the phone).



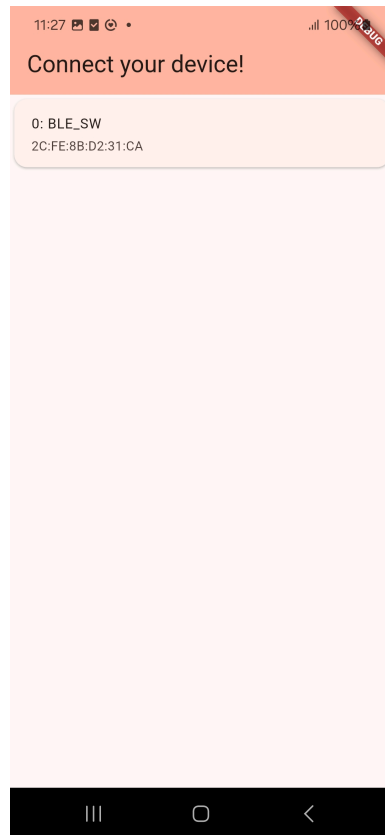
In the bottom-right corner, click on the text that shows the currently selected device (it might say Windows-x64 or No Device). A device selection menu will appear at the top of your screen. Select your phone (e.g., SM S911B, Pixel 7, ...) from the list.



6. Press "Start Debugging" (or run `flutter run` on terminal) to build and launch the app on your device. Make sure that

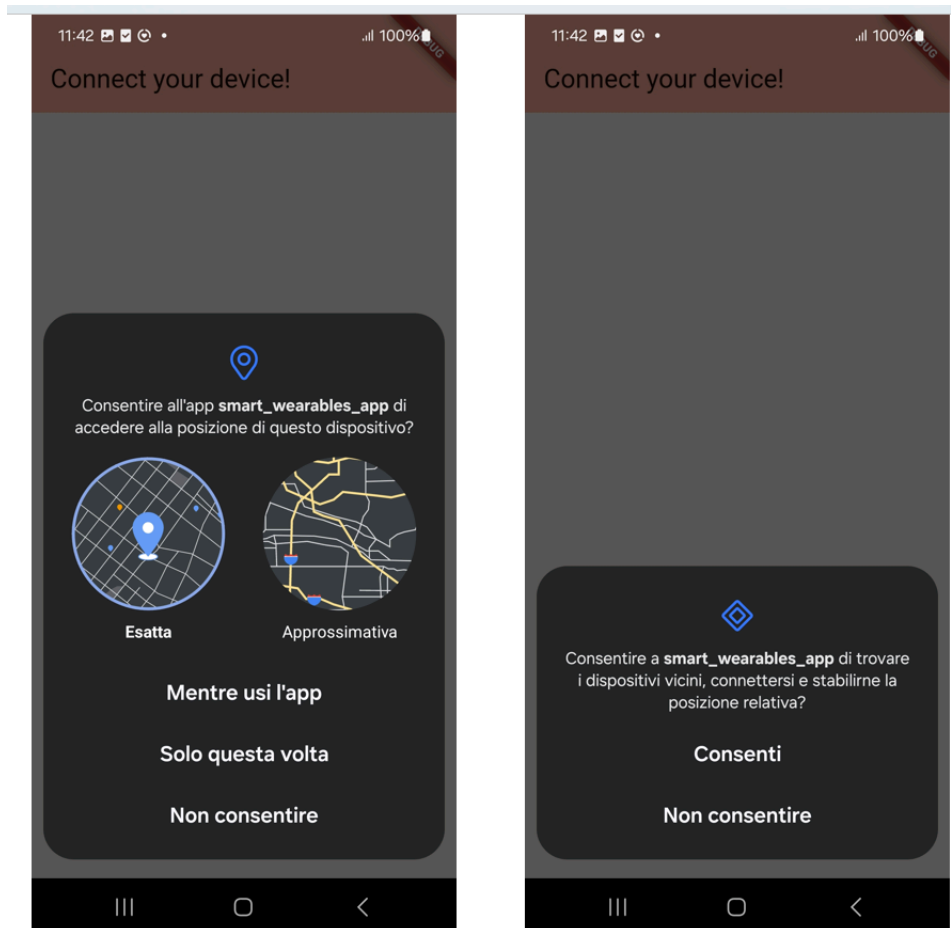


7. You should see the first homepage as shown below:

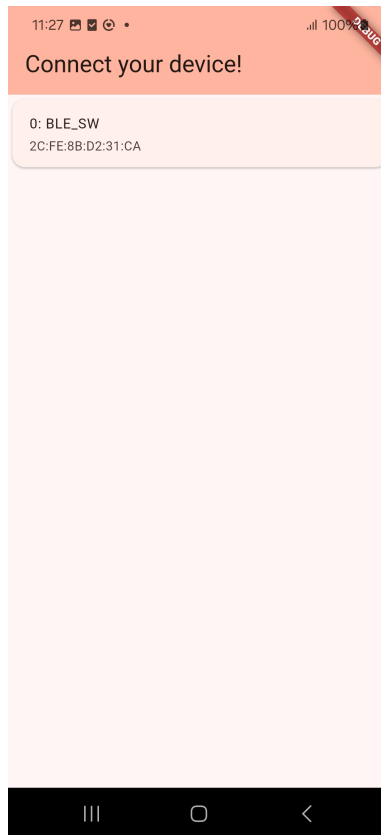


Section 2: First Run

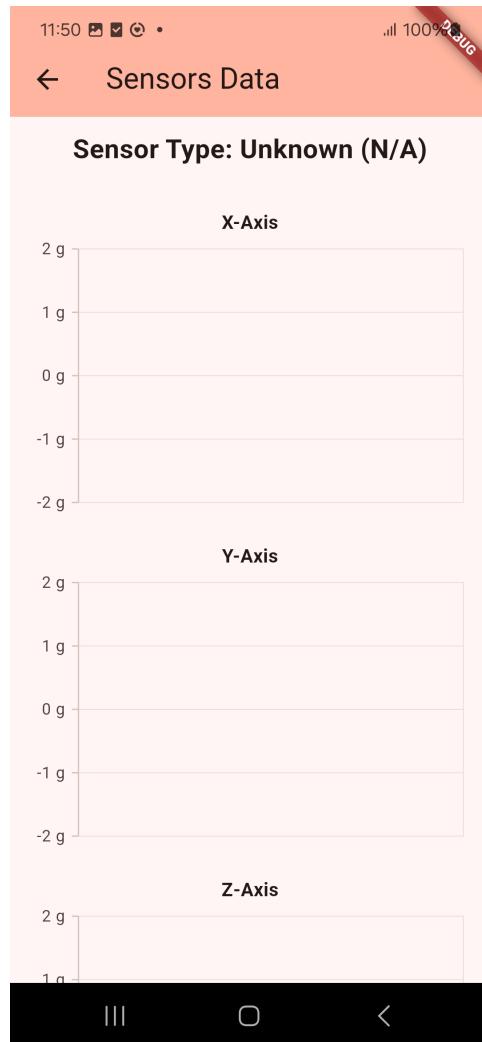
1. **Prepare the Board:** Power on the Course Development Board. Ensure the "MainBoard_IMU_Logger" firmware is running (Blue LED should be blinking).
2. **Launch the App:** Open the "Smart Wearables App" on your smartphone.
3. **Grant Permissions:** On the first launch, the app will ask for Bluetooth and Location permissions. You must allow both for BLE scanning to work.



4. **Scan & Connect:** The app will automatically scan for devices that match the firmware's Service UUID. Tap your device name (e.g., BLE_SW) in the list to connect.



5. **Start Data Stream:** After connecting, the app will navigate to the chart page. The graphs will be empty. Press the "USER BUTTON" on the physical board to begin the data acquisition and streaming.



6. **View Data:** The X, Y, and Z charts will immediately begin plotting the accelerometer data in real-time.

